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Data Analytics with R

Quality Laboratory Manual

Experiment No. 08

To implement and apply linear regression `lm()` to predict a continuous variable and interpret coefficients and plot regression lines



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Experiment No. 08

Title of Experiment: To implement and apply linear regression using `lm()` in R to predict a continuous variable, interpret coefficients and plot regression lines

Aim of Experiment: To build a linear regression model using R, predict a continuous dependent variable, interpret model coefficients and visualize the regression line

System Requirements –

- Operating System: Windows 10 or above
- RAM: Minimum 4 GB
- Hard disk: 10 GB free space
- Processor: 2.33 GHz or higher

Software/s Needed for Experiment

- R programming language (latest stable version)
- RStudio IDE
- Spreadsheet software (Microsoft Excel/ Libre Office)
- Built-in R graphics package
- Internet Connection (for dataset access)/ CSV dataset file

Experiment Objectives:

- To understand the concepts of linear regression and use `lm()` function
- Identify dependent and independent variables to interpret regression coefficients
- Evaluate model summary output
- Visualize regression line on scatter plot
- Predict values using regression model

Experiment Outcomes:

After completing the experiment, students will be able to –

- Build regression models in R
- Analyze relationships between variables
- Interpret slope and intercept
- Make predictions using trained models
- Visualize and validate regression results
- Apply regression in real-world analytics problems

Theory:

Linear Regression

Linear Regression is one of the most fundamental techniques in Machine Learning and statistics used to model the relationship between a dependent variable (response) and one or more independent

variables (predictors).

The goal of linear regression is to find a best-fit line that explains how the dependent variable changes with respect to independent variables.

The general equation of a simple linear regression model is:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Linear Regression in R using lm() Function

In R, linear regression is implemented using the built-in lm() (linear model) function.

Syntax:

model = lm(formula, data)

Where, model = name of regression model, formula = relationship between variables, data = dataset containing variables

Working Principle of lm()

The lm() function fits a linear model by estimating coefficients using the Ordinary Least Squares (OLS) method.

OLS minimizes the sum of squared differences between:

- Actual values
- Predicted values

Mathematically:

$$\text{Minimize } \sum (Y - \hat{Y})^2$$

This ensures the best possible line is fitted to the data.

Types of Linear Regression in R

a) Simple Linear Regression

Involves one independent variable: model = lm(y ~ x, data = dataset)

b) Multiple Linear Regression

Involves multiple independent variables: model = lm(y ~ x1 + x2 + x3, data = dataset)

Output Interpretation of lm()

After fitting the model: summary(model)

Model Prediction

Once the model is trained, predictions can be made using: predict(model, newdata)

Assumptions of Linear Regression

For accurate results, linear regression assumes:

- 1) Linearity – Relationship between variables is linear
- 2) Independence – Observations are independent
- 3) Homoscedasticity – Constant variance of errors
- 4) Normality – Errors are normally distributed

Task to Perform**Modeling Fuel Consumption Based on Vehicle Speed using Linear Regression**

An automobile company wants to analyze how vehicle speed affects fuel consumption. Engineers believe that fuel consumption changes with speed and want to build a predictive model.

As a data analyst, you are required to –

- 1) Analyze the relationship between speed and fuel consumption
- 2) Build a regression model
- 3) Predict fuel consumption at a given speed
- 4) Provide insights for optimal driving efficiency

Dataset –

```
vehicle_data <- data.frame(
  Speed =
c(20,22,25,28,30,32,35,38,40,42,45,48,50,52,55,58,60,62,65,68,70,72,7
5,78,80,82,85,88,90,92,95,98,100,102,105,108,110,112,115,118,120,122,
125,128,130,132,135,138,140,145
),
  Fuel_Consumption =
c(5.2,5.3,5.5,5.7,5.8,6.0,6.1,6.3,6.5,6.6,6.8,7.0,7.1,7.3,7.5,7.7,7.9
,8.0,8.3,8.5,8.7,8.9,9.2,9.4,9.6,9.8,10.1,10.4,10.6,10.8,11.0,11.3,11
.5,11.7,12.0,12.3,12.5,12.8,13.1,13.4,13.6,13.9,14.2,14.5,14.8,15.1,1
5.5,15.8,16.2,16.8))
```

Note: Consider the speed unit in km/h and fuel consumption in litres per 100 km

Task-1 – Data Understanding

- Display the dataset
- Identify –
 - Independent variable
 - Dependent variable

Task-2 – Build Linear Regression Model

Use lm() –

Dependent variable for fuel_consumption
Independent variable for speed

Task-3 – Model Summary

- Display summary(model0)
- Identify –
 - Intercept
 - Slope
 - R-squared

Task-4 – Visualization

- Plot scatter plot
- Add regression line

Task-5 – Prediction

Predict fuel consumption at speed = 95 km/h

Task-6 – Interpretation

Answer the following –

- 1) How does speed affect fuel consumption?
- 2) Is the relationship linear?
- 3) At what speed range is fuel efficiency better?

Conclusion:

The experiment successfully demonstrated the implementation of linear regression using R. students learned how to build a model using `lm()`, interpret coefficients, evaluate model performance and visualize regression lines. Linear regression is a powerful and widely used technique for predictive analytics in real-world scenarios

Expected Oral Questions:

- 1) What is linear regression?
- 2) What does `lm()` function do?
- 3) What is slope and intercept?
- 4) What is R-squared?
- 5) What is dependent and independent variable?
- 6) What is least squares method?
- 7) How do you plot regression line?
- 8) What is residual?
- 9) How do you make predictions?

FAQs in Interview:

- 1) What does R-squared indicate?
- 2) What is overfitting?
- 3) What are assumptions of linear regression?
- 4) What happens if data is not linear?
- 5) Why is visualization important in regression?